Pomona College

Computer Science

Spring 2025: CSCI 181RT

Real-Time Systems in the Real World

Lecture 7

Tuesday, February 11, 2025
Edmunds Hall 105
2:45 PM - 4:00 PM

Professor Jennifer DesCombes

Agenda

- Go Backs
- Discussion on Reading
- Documenting Processing Flow/Sequence
- Initial Discussions of Tasks
- Lab Preview
- Assignment
- Look Ahead
- Action Items

Go Backs

- General?
- Lab Discussion
- Action Item Status
  - AI250204-1: malloc vs calloc - OK to Close?
    - void *malloc (size_t n)

      Returns a pointer to ’n’ bytes of un-initialized storage, or NULL if the request can not be satisfied.
    - void *calloc (size_t n, size_t size)

      Returns a pointer to ’n’ objects of the specified ‘size’ of initialized to zero storage, or NULL if the request can not be satisfied.

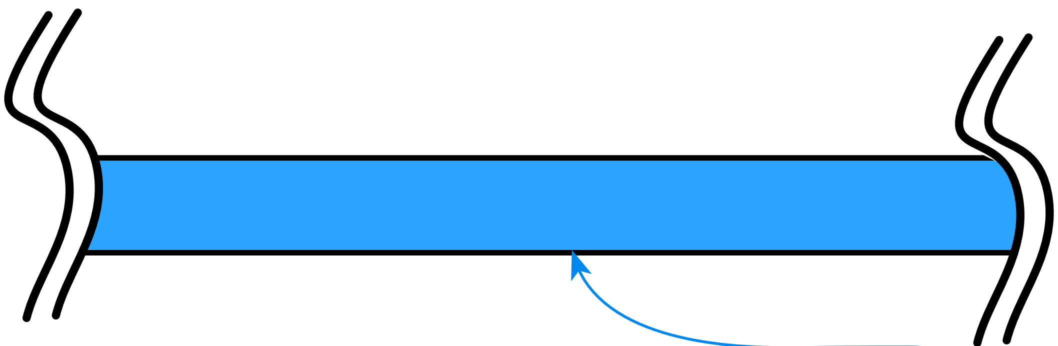
      Example: myStructListPtr = calloc (100, sizeof(myStructType));
  - AI250206-1: Find reasonable article/reference on microcode
  - AI250206-2: Jackson earns 1% bonus for catching error in predecrement and post decrement

Concepts Discussed In Lectures 2 - 6

- Communications - Verbal
- Requirements Analysis and Complexity
- Simple Polled Real-time System
- Introduction to Stack Utilization
- Introduction to Computer Architecture
- Interrupt Processing and More on Stacks

A Simple Real-Time System Design
Elevator Music Controller

main ( )



```
.
.
bool button1Pressed;
bool button2Pressed;
bool button3Pressed;
bool button4Pressed;

void readButtons( void );
void processButtons ( void );
.
.
  while(true)
  {
    readButtons( );
    processButtons( );
  }
.
.
```

Documenting Processing Flow/Sequence

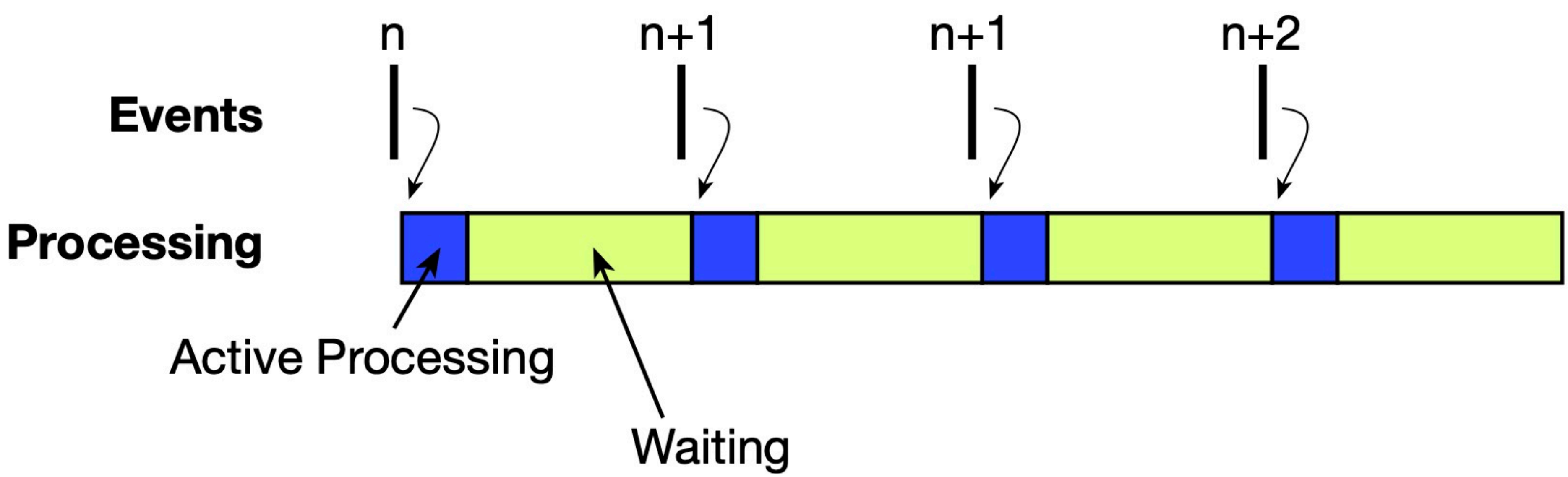
- How to Discuss Complex Processing Sequences
- How to Document Complex Processing Sequences
  - Simple Cases - Synchronous
  - Complex Cases - Asynchronous or Multi-synchronous

Discussing Complex Processing Sequences

- Why
  - Share With Other Designers
  - Present to Managers or Customers
- Clear Representation of What is Happening / Planned
  - Will Receive Better Feedback
  - Will Reduce Questions
  - Allows Tailoring of Presented Data

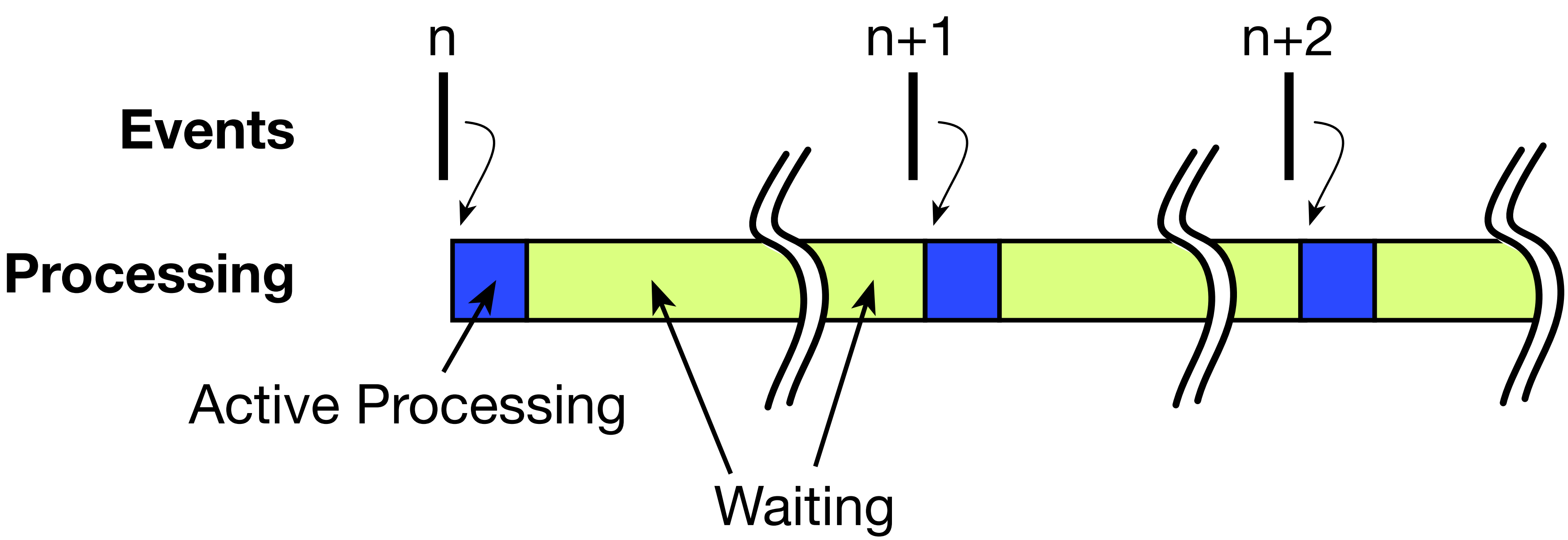
Processing - Simple Cases - Synchronous

- Graphically - 1 Image is Worth 100 Words (Inflation)
  - Event Marks
  - Processing BarsGraphically - 1 Image is Worth 100 Words
- Repetition at a Fixed Rate
- Events and Processing are Locked to Each Other



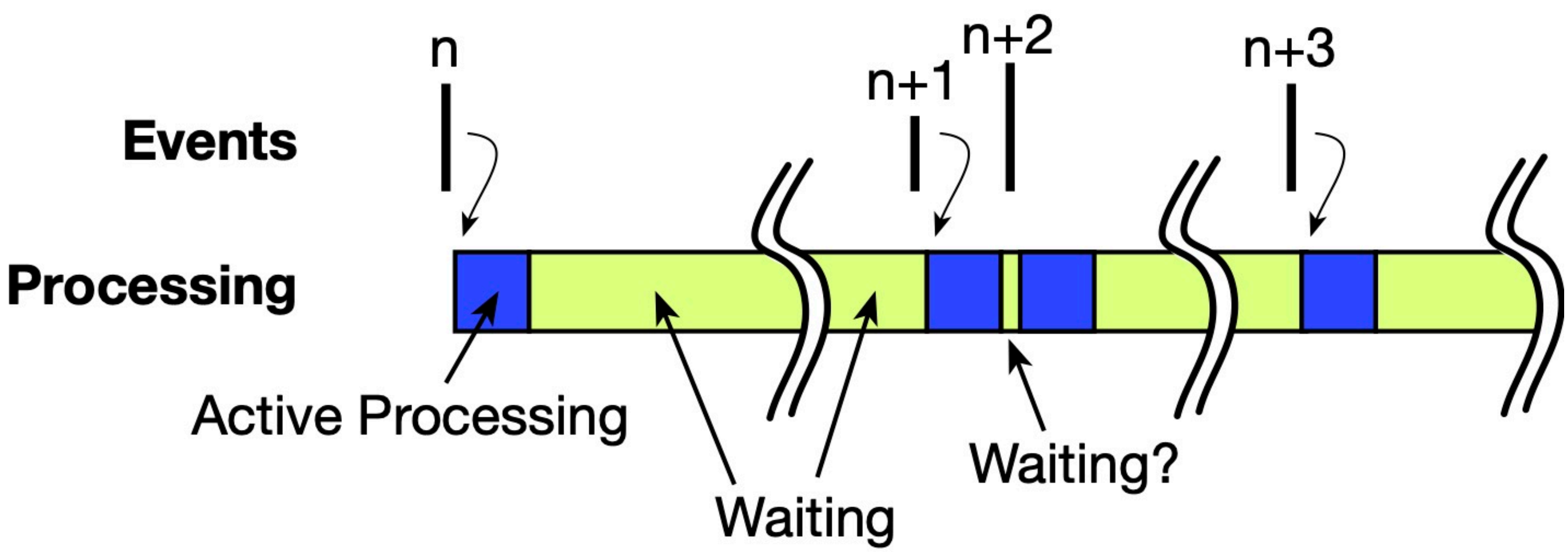
Documenting Complex Processing Sequences

- Example - Asynchronous
- Events and Processing are Locked to Each Other
  - An Event Occurs (n, n+1, n+2, etc.)
  - Processing of the Event Follows



Processing - Complex Cases - Asynchronous

- Repetition at a Variable Rate
- Repetition of More Than One Event at Different Rates
- Events and Processing are Still Typically Locked to Each Other



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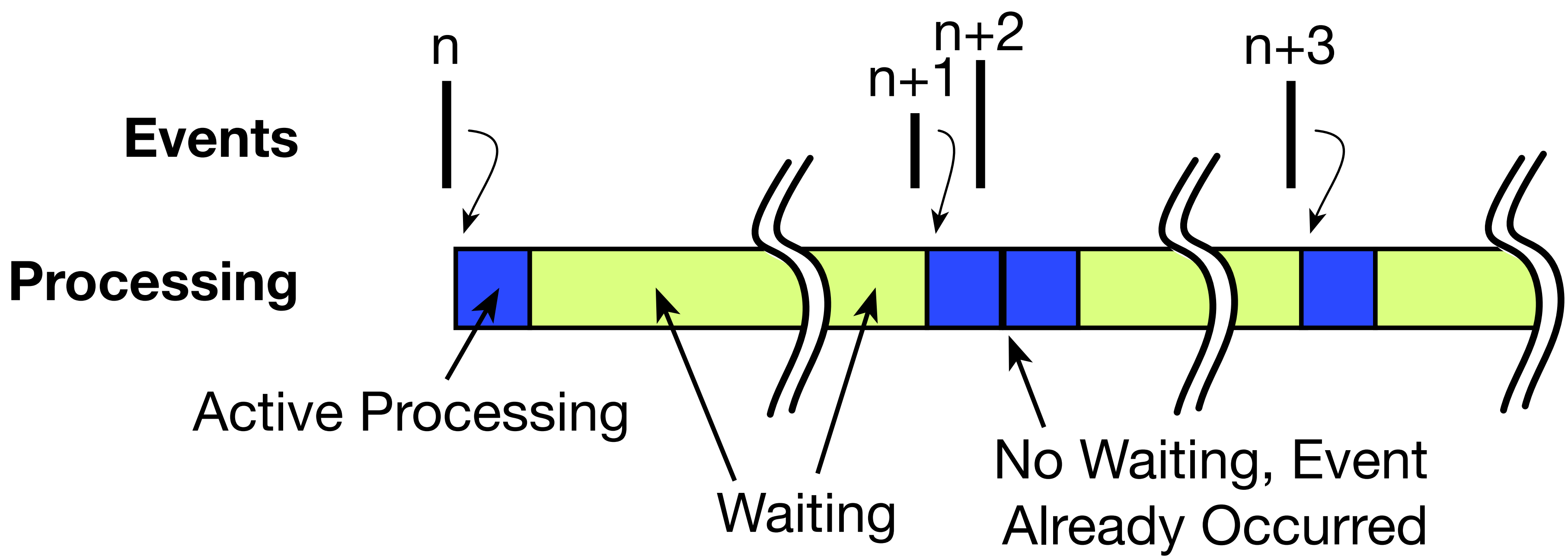
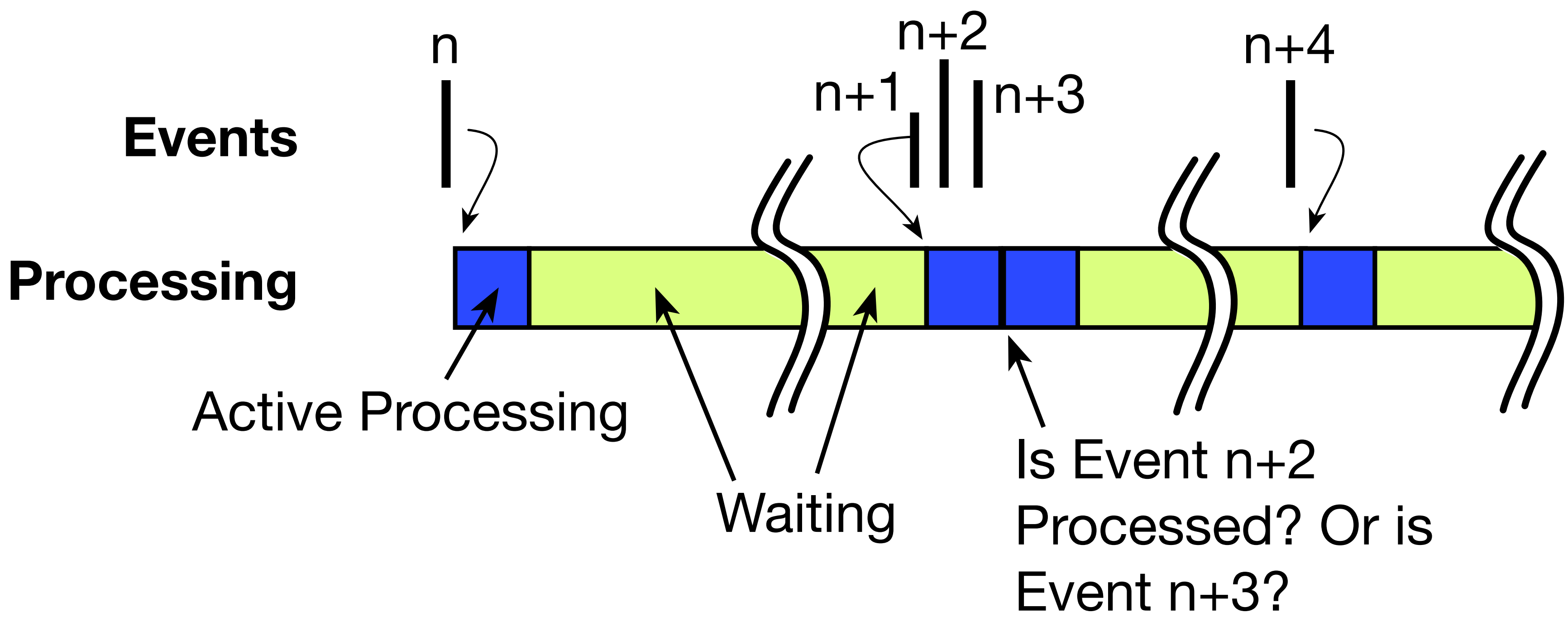
Processing - Complex Cases - Asynchronous

- Overlap of Event and Processing - n+1 and n+2
- Processing of n+1 Continues during Event n+2
- Might Be OK?



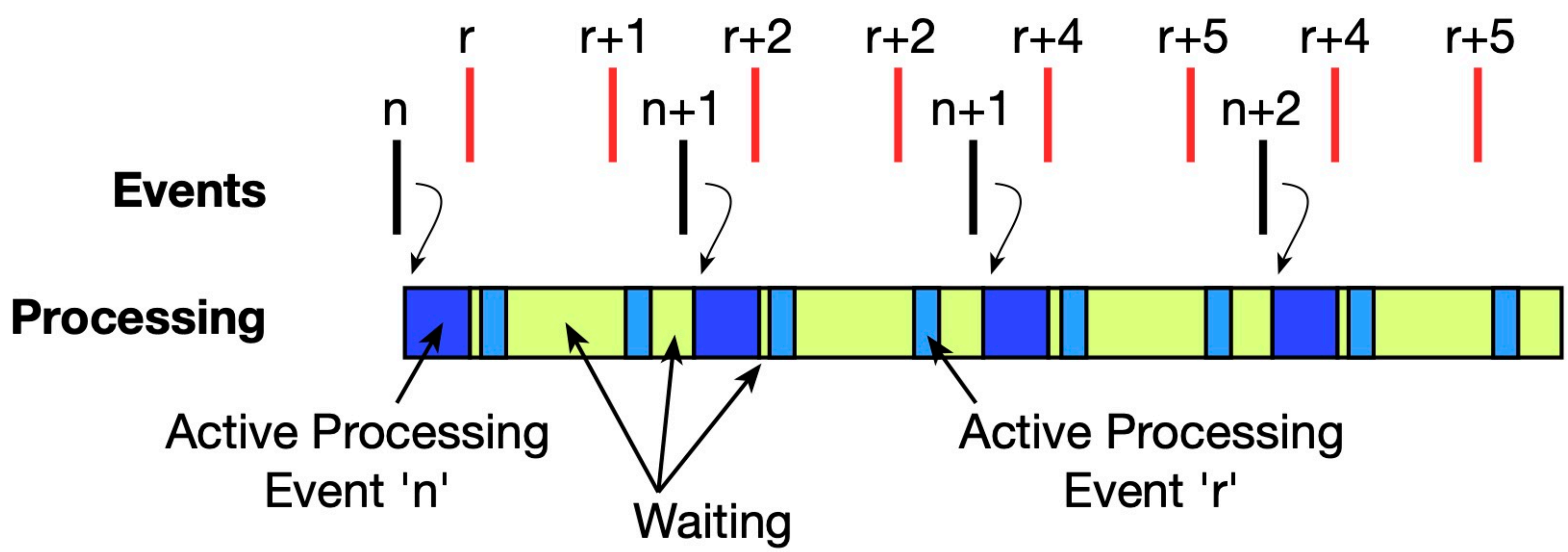
Processing - Complex Cases - Asynchronous

- Overlap of Event and Processing - n+1, n+2, and n+3
- Processing of n+1 Continues during Event n+2 and n+3
- Might Be OK? [Editors Note: Probably not]



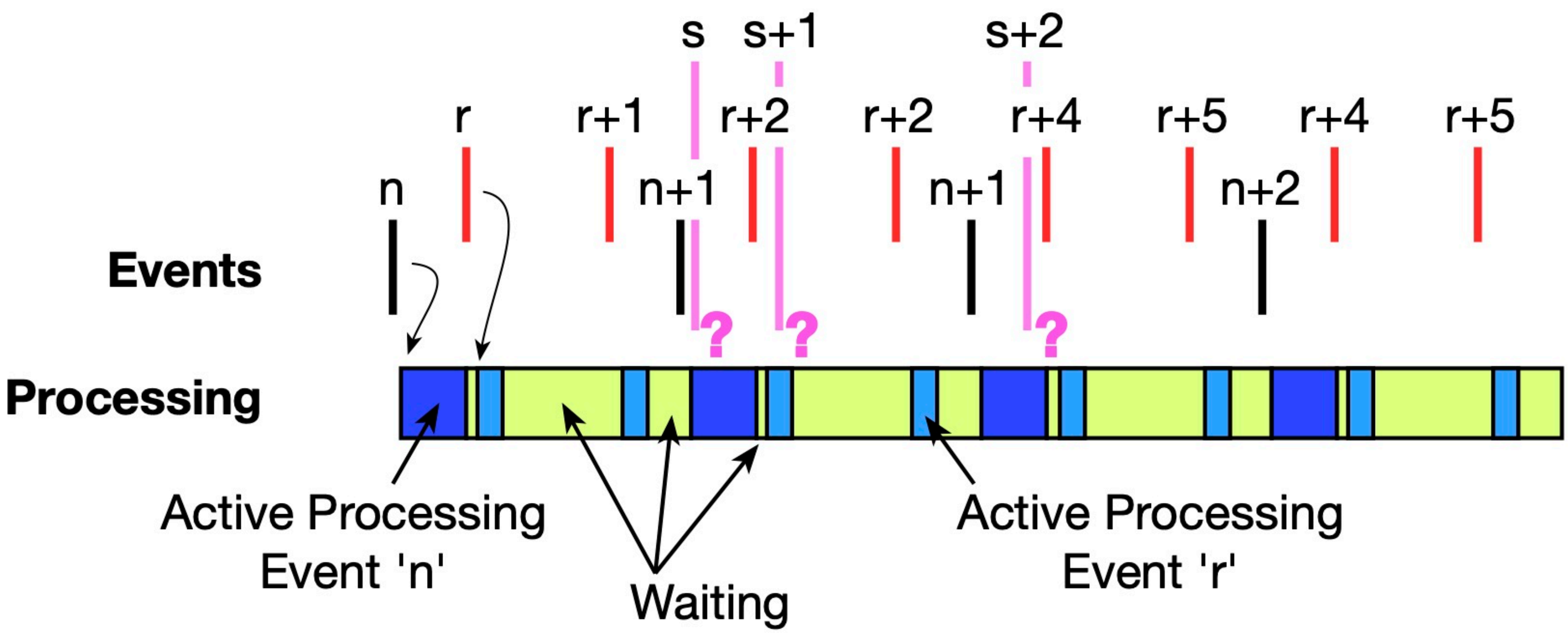
Processing - Complex Cases - Multi-synchronous

- Repetition at a Variable Rates
- Repetition of More Than One Group of Events at Different Rates
- Events and Processing May be Locked to Each Other
- Events and Processing May be Locked to Multiple Events



Processing - Complex Cases - Fully Mixed

- Everything at Once
- Issues?



Initial Discussions of Tasks

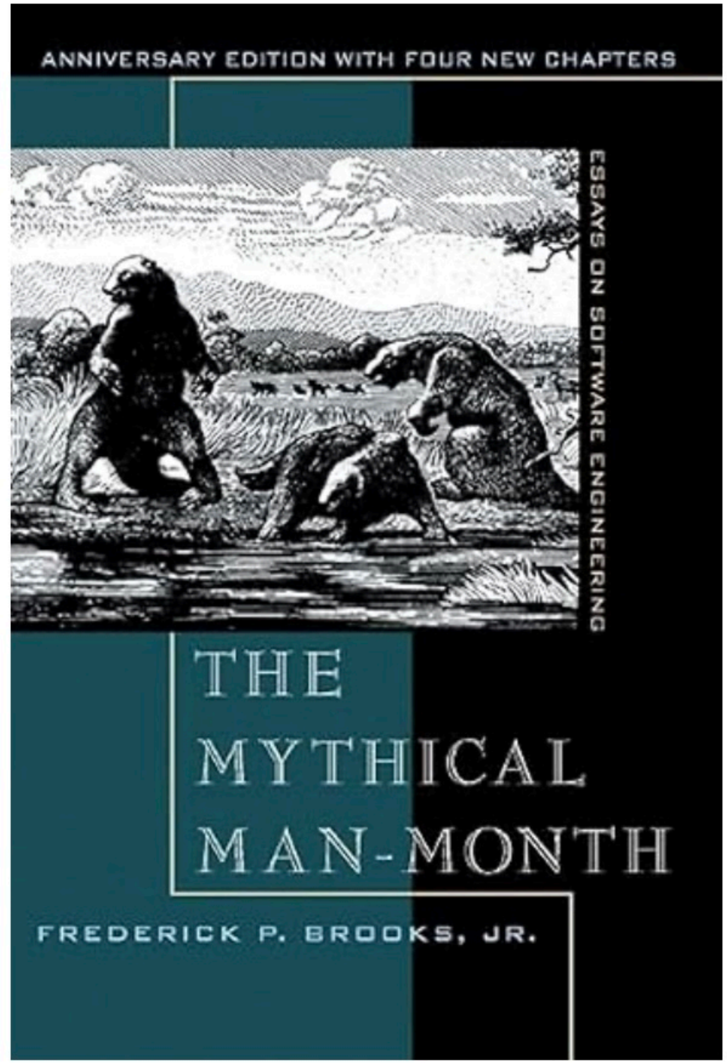
- Why Create a Task
- What Defines a Task

Lab Preview

- Evaluation Boards
- USB-micro to USB-A Cables
- Software Installation - Mostly Achieved
- Compilation of Micro-controller Software
- Use of Virtual Comm Window
- Software/Project Structure
  - Files and Folders
  - Logical and Physical

Assignment - Readings

- The Mythical Man-Month
  - Chapter 2: The Mythical Man-month
- Send Me Discussion Topics by 10:00 on Thursday.



Look Ahead

- Discussion of Reading
- Discussion of Lab 2
- More Discussions of Simple Task Design

Action Items and Discussion

AI#:	Owner	Slide #	Document	Action